

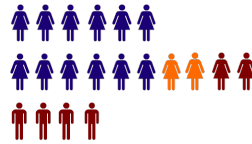
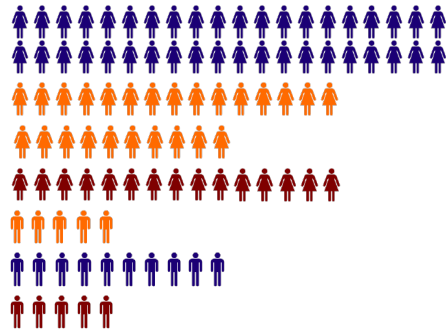


Survey Methods

2.3 - Sampling strategies

- Probability vs non-probability samples
- Sampling strategies
- Systematic sampling and systematic error
- Sampling strategies and data analysis

Sampling & error



Lack of
Representativeness

Systematic

If we had to repeat
the sampling, we will
commit the same
error (in the
estimates)

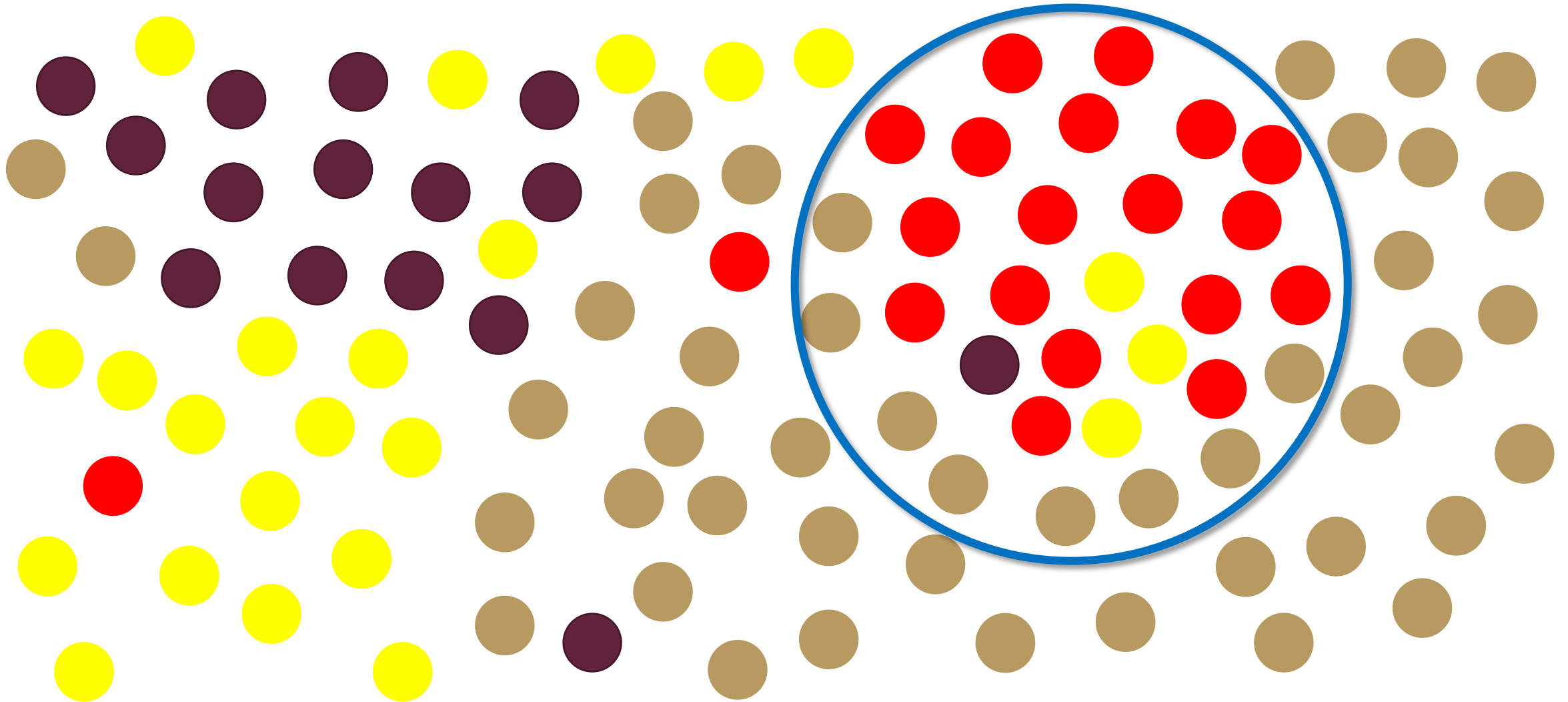
It is not affected by
sample size

Random

If we had to repeat
the sampling, we will
obtain different results

It tend to decrease
with increasing
sample size

Looking in the wrong place....



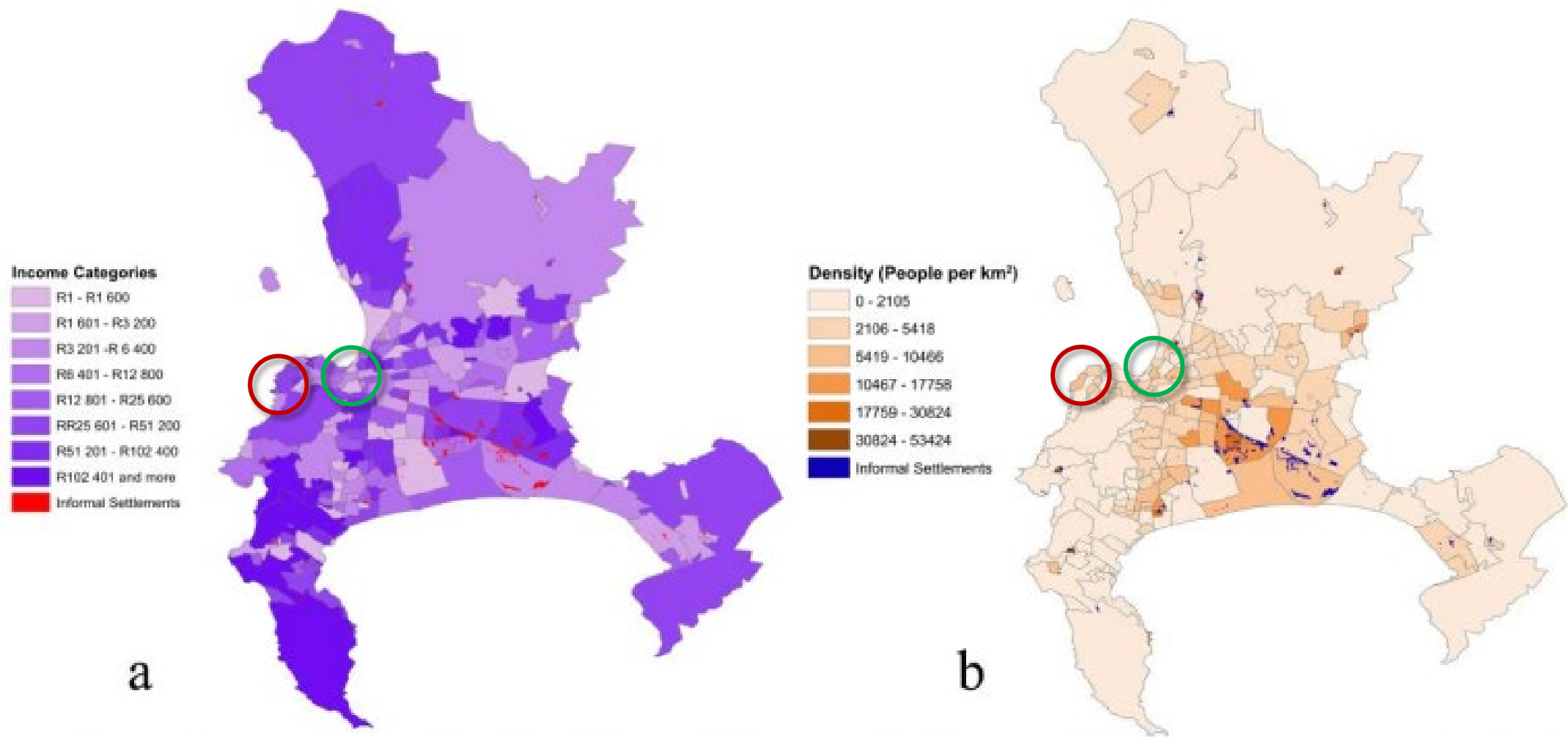
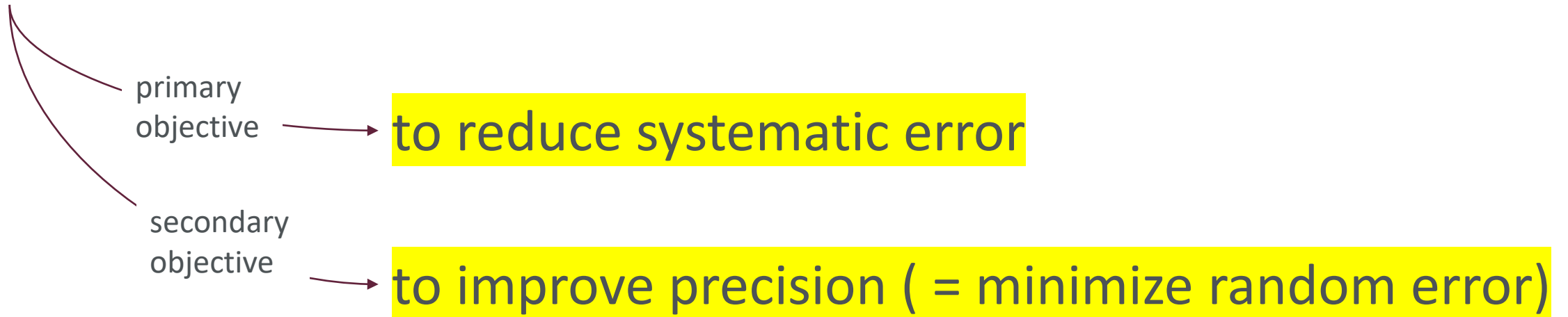


Figure 1: Cape Town suburbs delineated by monthly household income categories (a) and population density (b), overlaid with presence of informal settlements.

Data sourced from City of Cape Town (2016a)

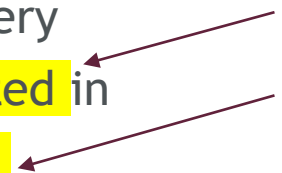
(Currie, Musango, and May 2017)

Sampling strategies = how to select a sample



Probability vs non-probability samples

A **probability sampling** strategy is one in which every unit in the population **has a chance of being selected** in the sample, and this probability **can be accurately determined**.



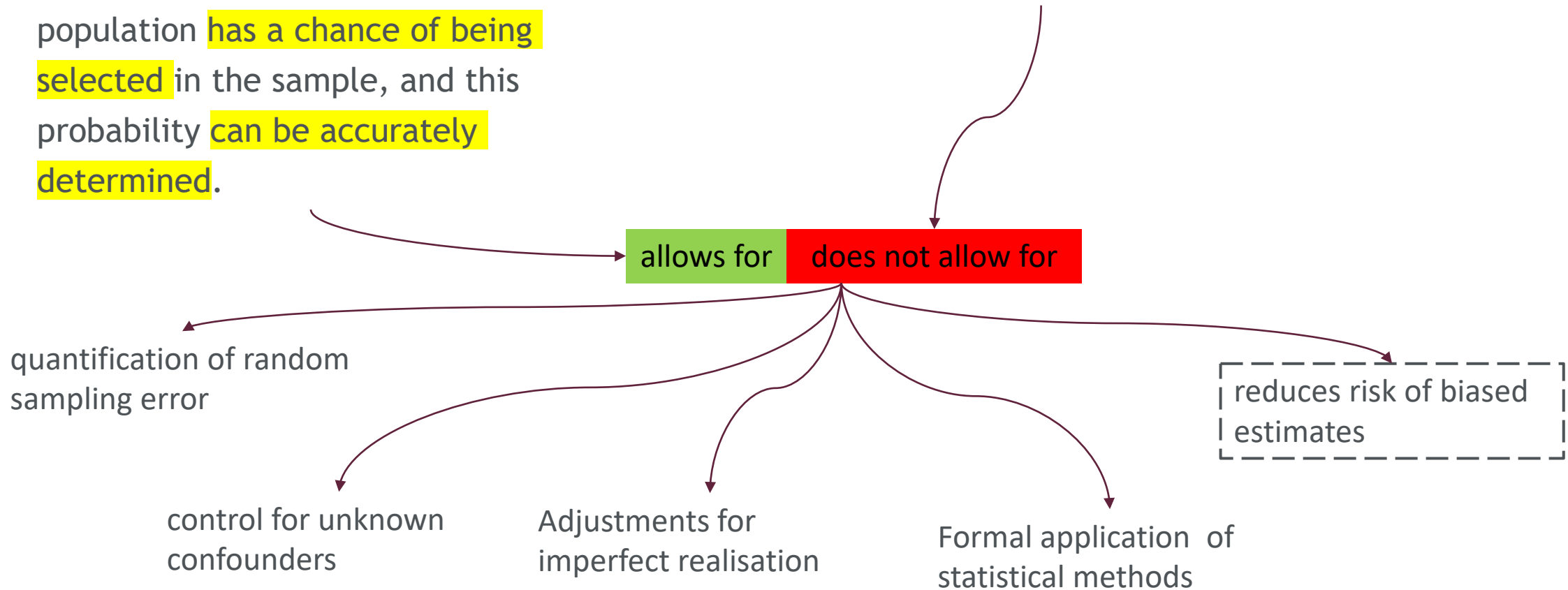
probability of selection $p > 0$
 p is known

Everything else is **non-probability sampling**

Advantages of probability sampling

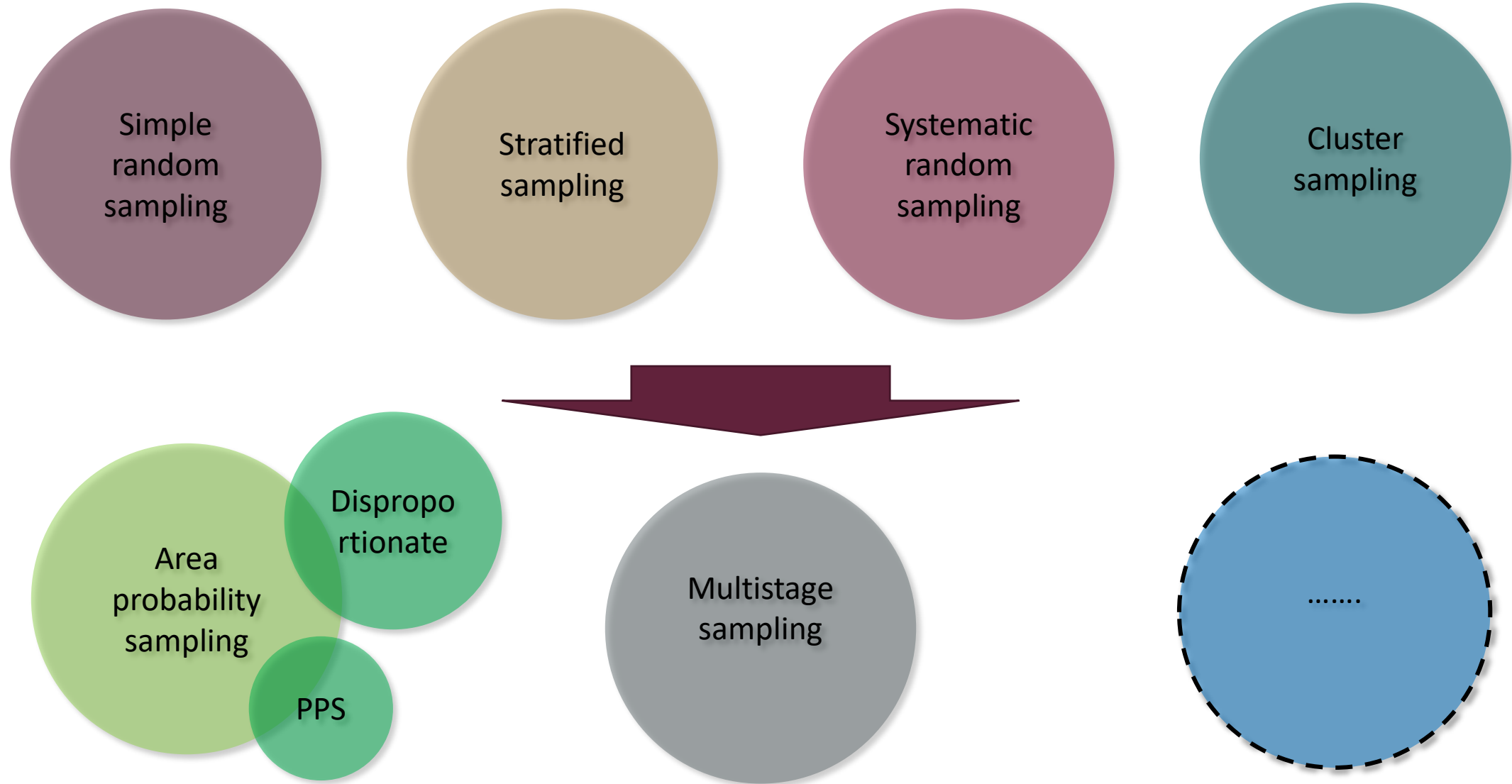
A **probability sampling** strategy is one in which every unit in the population **has a chance of being selected** in the sample, and this probability **can be accurately determined**.

Everything else is **non-probability sampling**

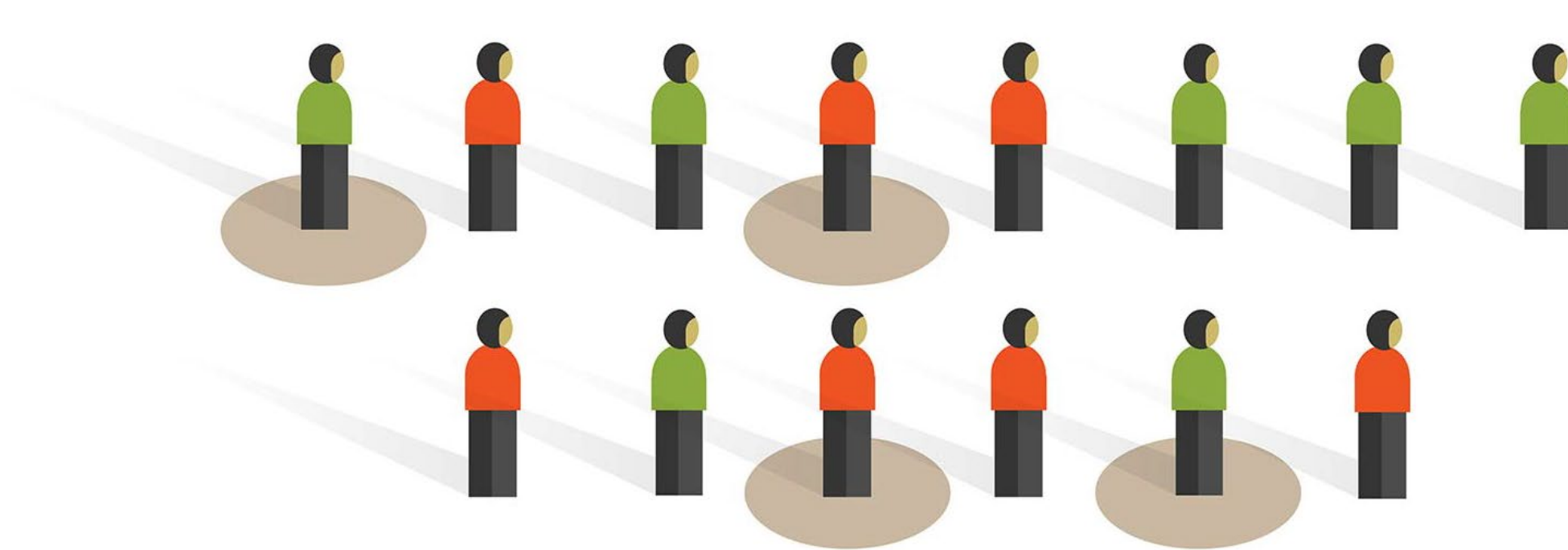




Probability sampling strategies



Simple random sampling



Random selection from a sampling frame

Table of Random Numbers

36518	36777	89116	05542	29705	83775	21564	81639	27973	62413	85652	62817	57881
46132	81380	75635	19428	88048	08747	20092	12615	35046	67753	69630	10883	13683
31841	77367	40791	97402	27569	90184	02338	39318	54936	34641	95525	86316	87384
84180	93793	64953	51472	65358	23701	75230	47200	78176	85248	90589	74567	22633
78435	37586	07015	98729	76703	16224	97661	79907	06611	26501	93389	92725	68158
41859	94198	37182	61345	88857	53204	86721	59613	67494	17292	94457	89520	77771
13019	07274	51068	93129	40386	51731	44254	66685	72835	01270	42523	45323	63481
82448	72430	29041	59208	95266	33978	70958	60017	39723	00606	17956	19024	15819
25432	96593	83112	96997	55340	80312	78839	09815	16887	22228	06206	54272	83516
69226	38655	03811	08342	47863	02743	11547	38250	58140	98470	24364	99797	73498
25837	68821	66426	20496	84843	18360	91252	99134	48931	99538	21160	09411	44659
38914	82707	24769	72026	56813	49336	71767	04474	32909	74162	50404	68562	14088

STATA

```
gen x = runiform(1,100)
```



```
x <- runif(1,1,100)
```

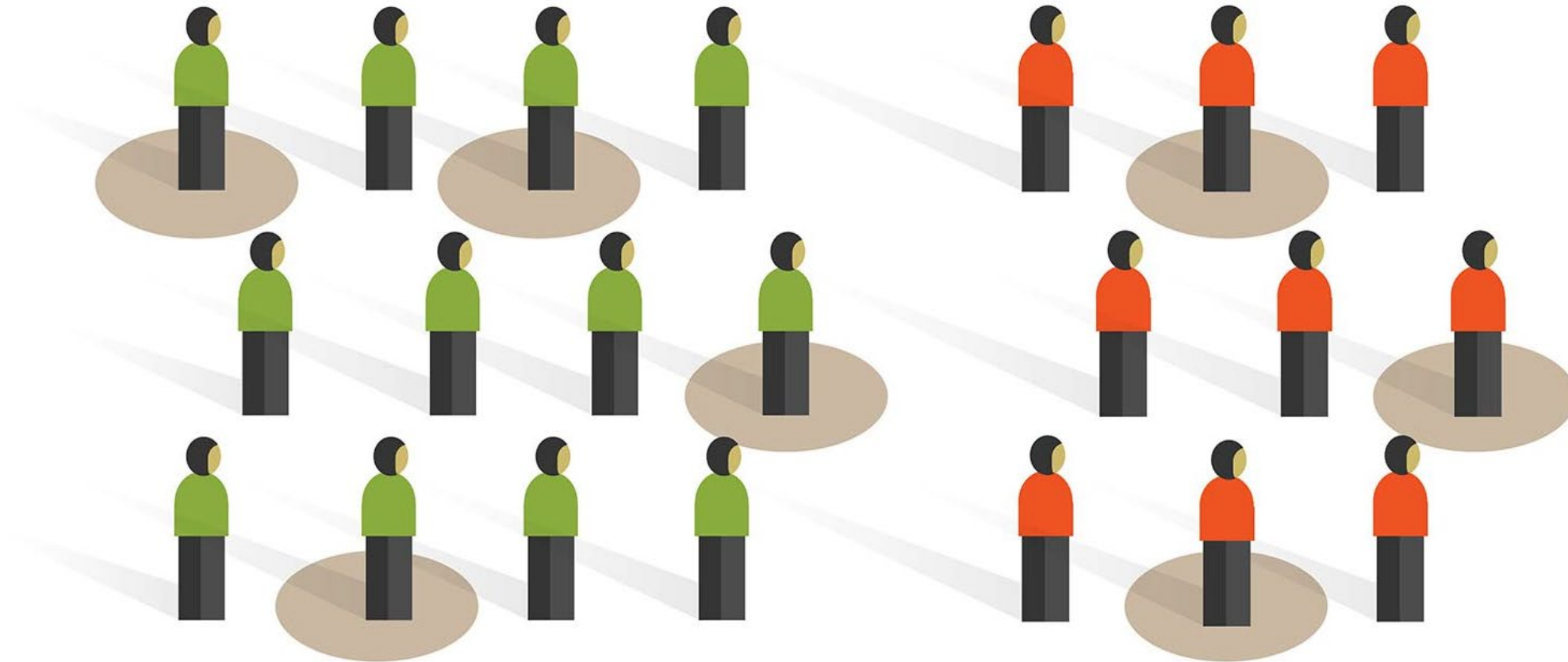


RAND()

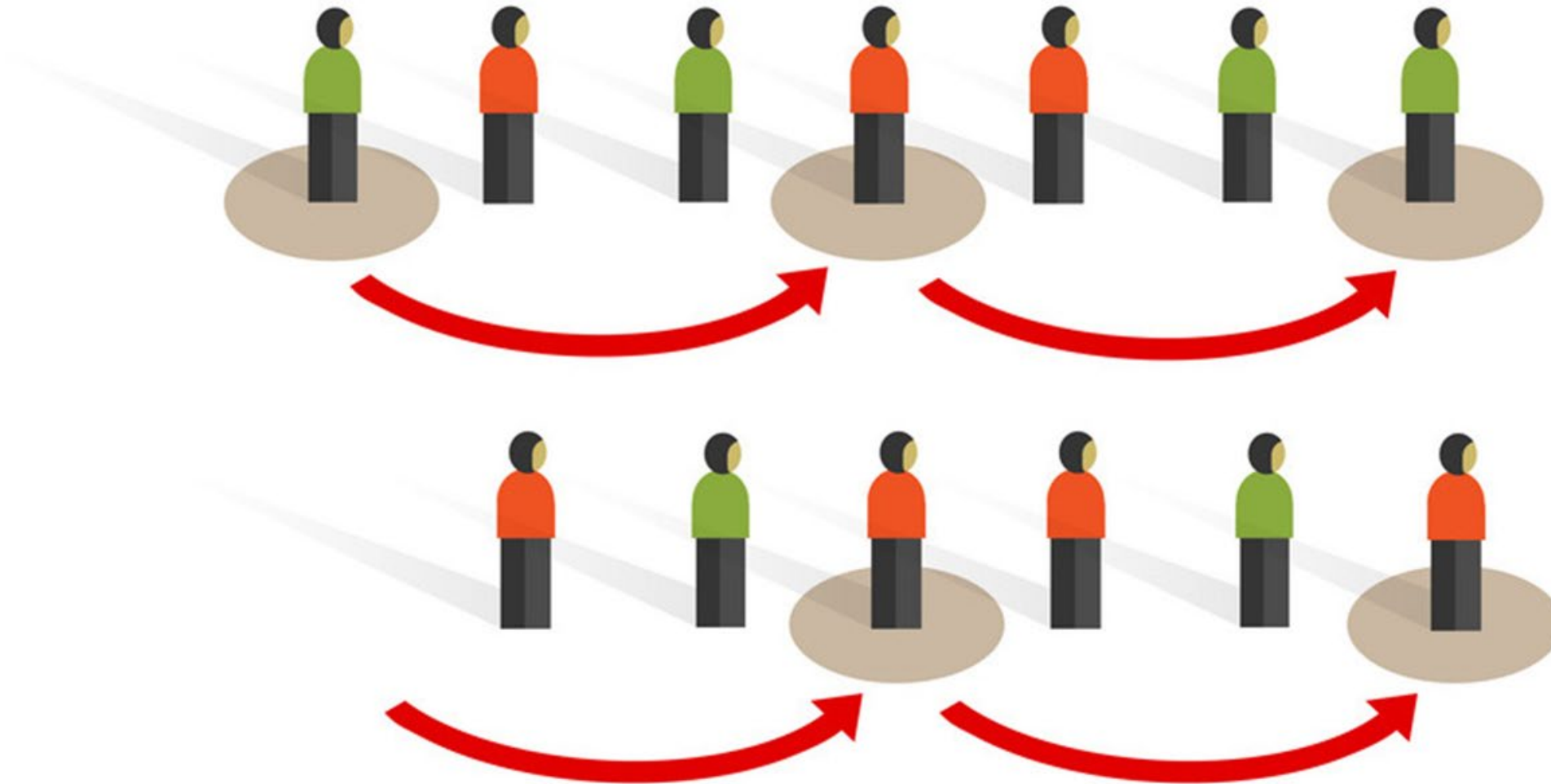


EXAMPLE

Stratified random sampling

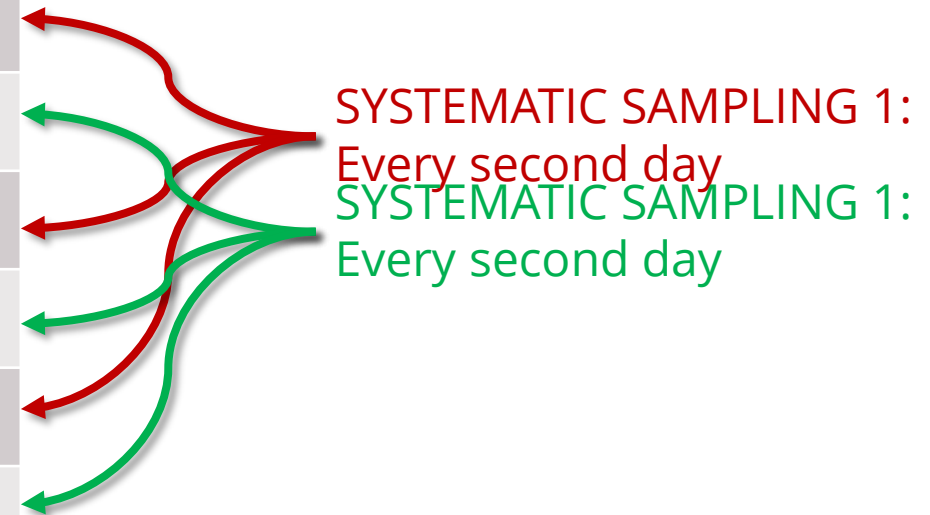


Systematic random sampling

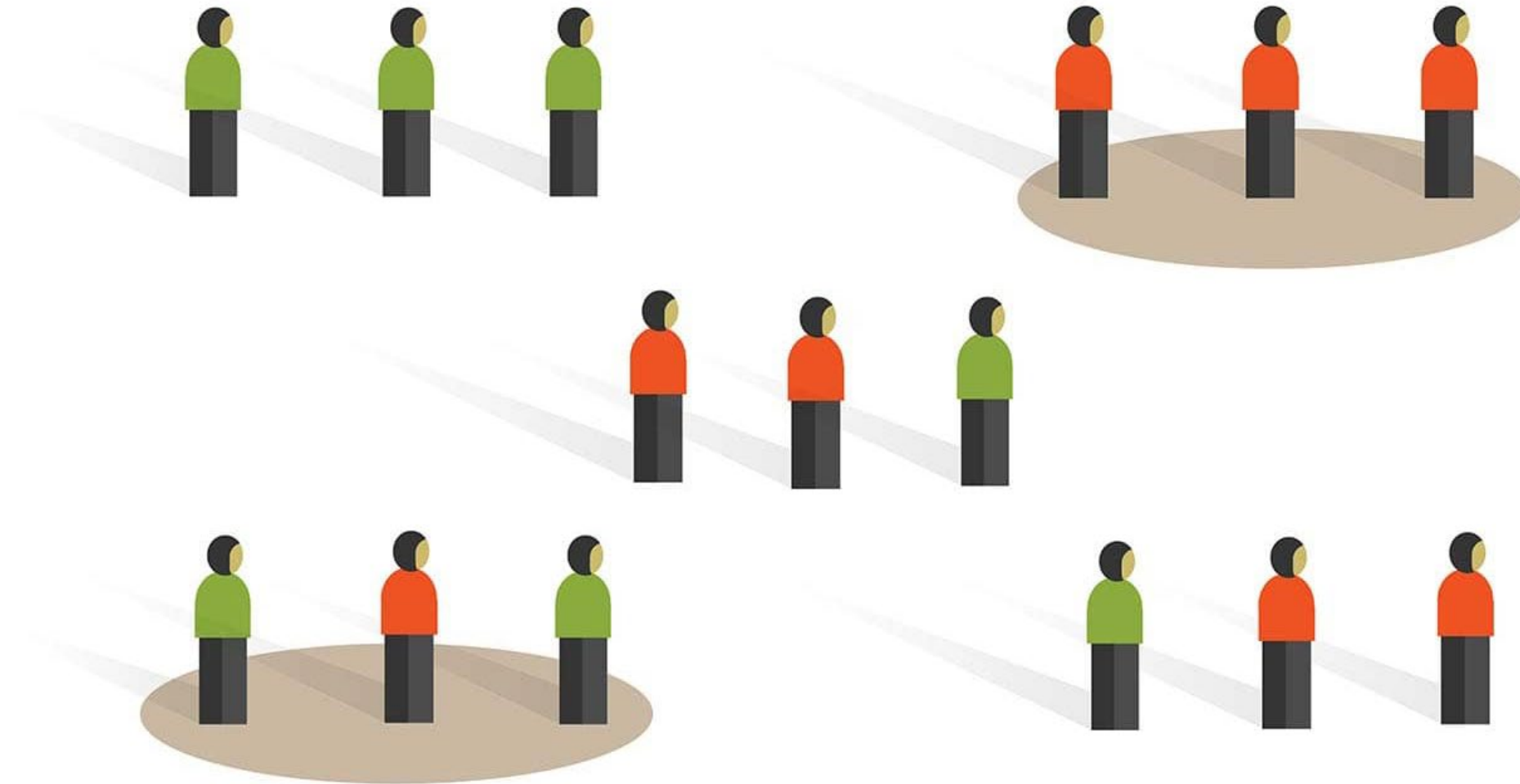


Diabetic clinic

Day	Type of patients
Monday	New patients
Tuesday	Returning patients
Wednesday	Returning patients
Thursday	Returning patients
Friday	Returning patients
Saturday	Returning patients

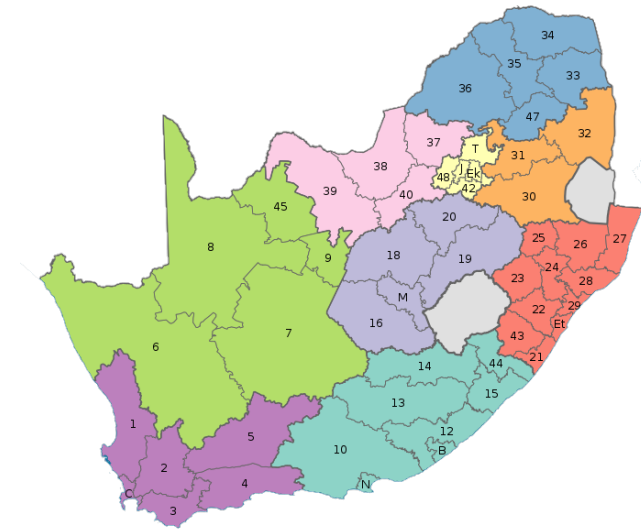


Cluster random sampling



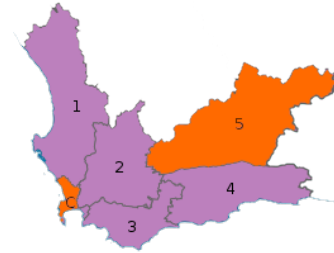
Multistage sampling

Stratification by province



Randomly select 2
districts in each province

Stage 1



Randomly select 3 clinics
in each district

Stage 2



Randomly select 4
patients in each clinic

Stage 3



Random selection
(not all clusters are included in the sample)

Large number

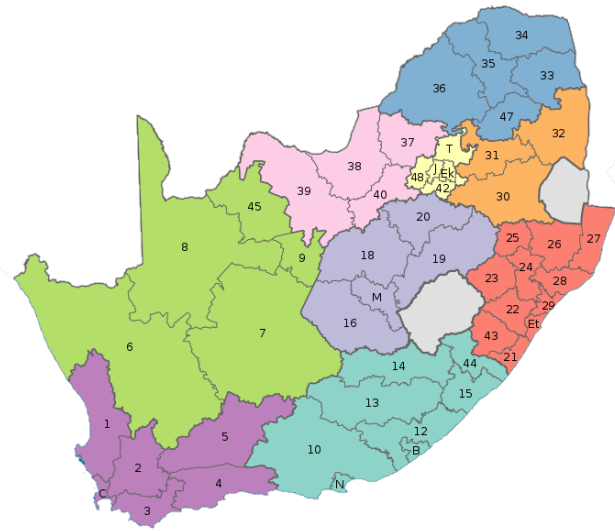
Cluster vs stratum

Non-random selection
(all strata are included in the sample)

Small number

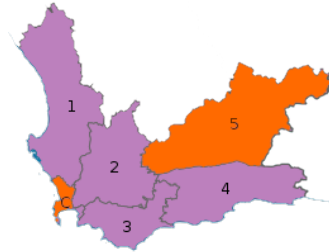
Multistage sampling

Stratification by province



Randomly select 2 districts in each province

Stage 1



Randomly select 3 clinics in each district

Stage 2



Randomly select 4 patients in each clinic

Stage 3



Area probability sampling

PPS

Disproportionate

Advantages and disadvantages

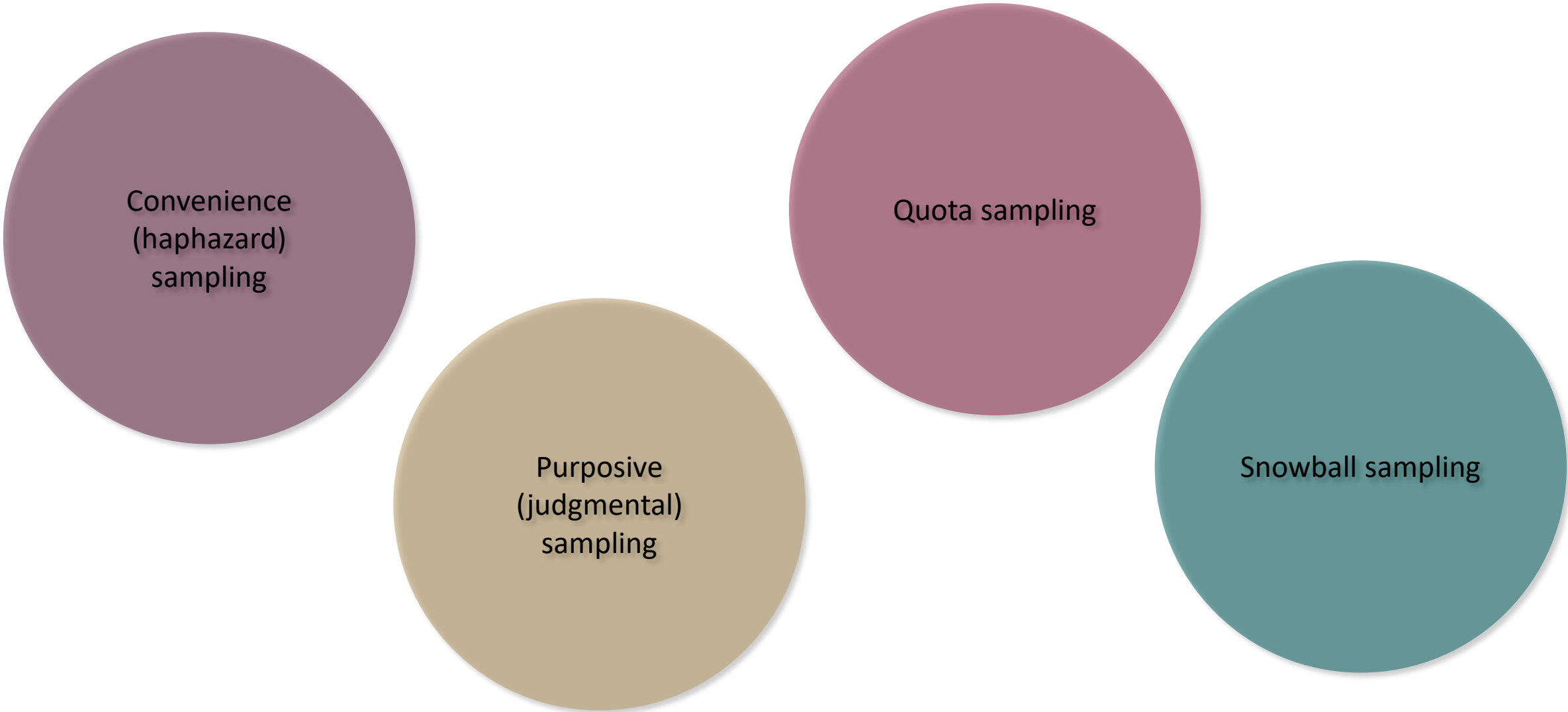
Design	Advantages	Disadvantages
Simple random	• Requires little knowledge of population in advance.	• May not capture certain groups of interest. • May not be very efficient.
Systematic	• Easy to analyze data and compute sampling (standard) errors. • High precision.	• Periodic ordering of elements in sample frame may create biases in the data. • May not capture certain groups of interest. • May not be very efficient.
Stratified	• Enables certain groups of interest to be captured. • Enables disproportionate sampling and optimal allocation within strata. • Highest precision.	• Requires knowledge of population in advance. • May introduce more complexity in analyzing data and computing sampling (standard) errors.
Cluster	• Lowers field costs. • Enables sampling of <i>groups</i> of individuals for which detail on individuals themselves may not be available.	• Introduces more complexity in analyzing data and computing sampling (standard) errors. • Lowest precision.

Efficiency

Risk of bias

Type of analysis

non-probability sampling strategies



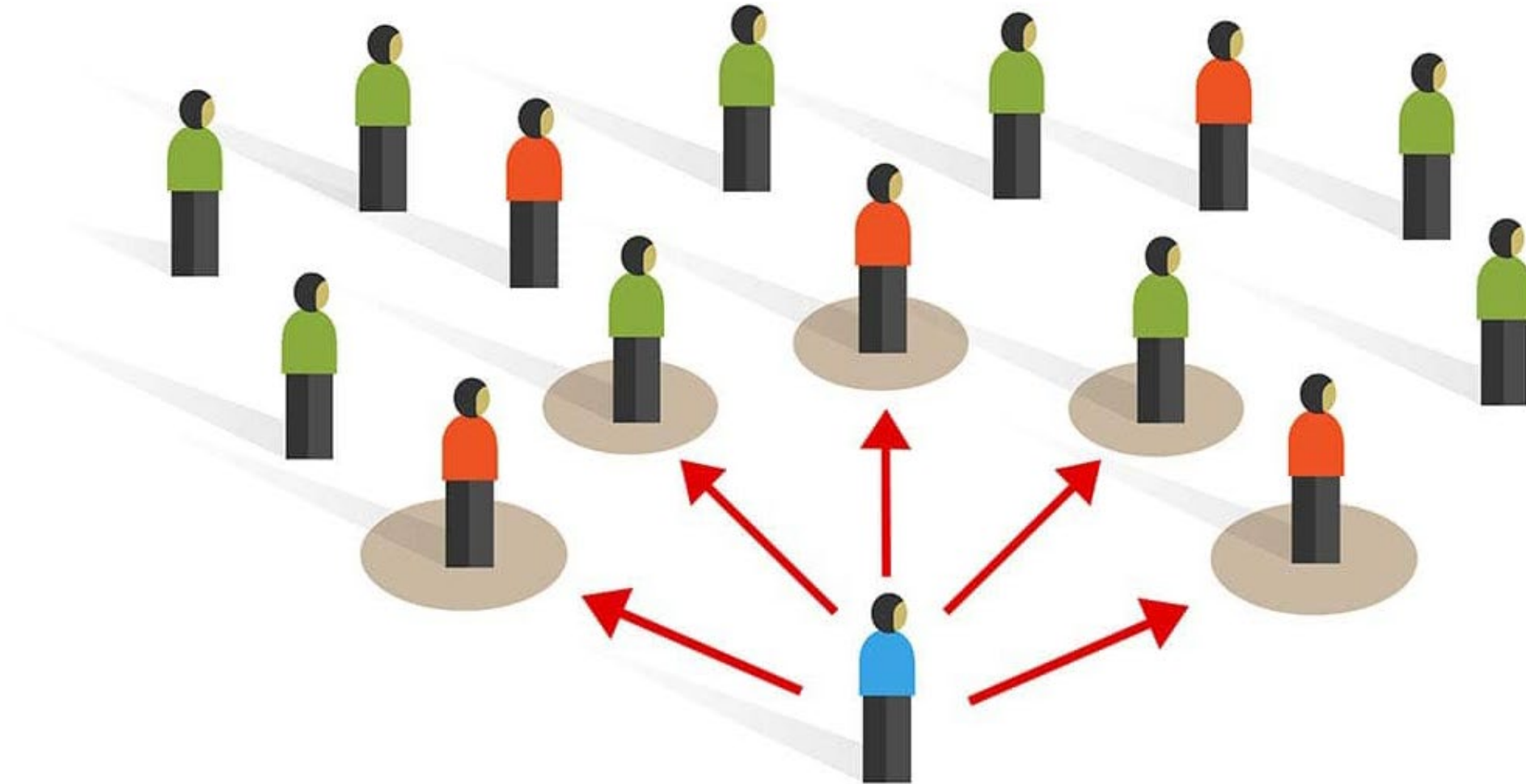
Convenience
(haphazard)
sampling

Quota sampling

Purposive
(judgmental)
sampling

Snowball sampling

Convenience sampling



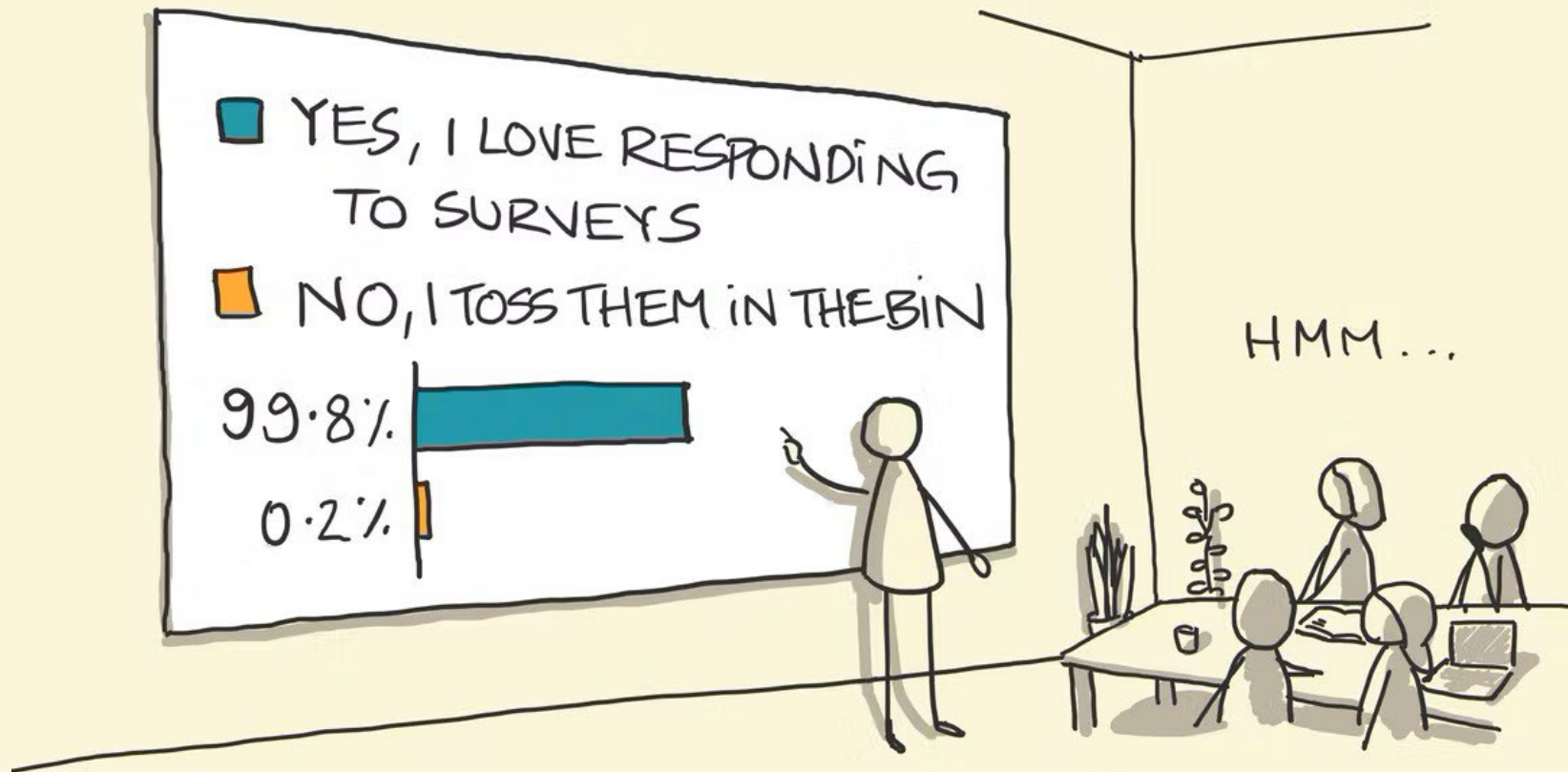
LOOKING UNDER THE LAMPOST



LOST YOUR
KEYS?

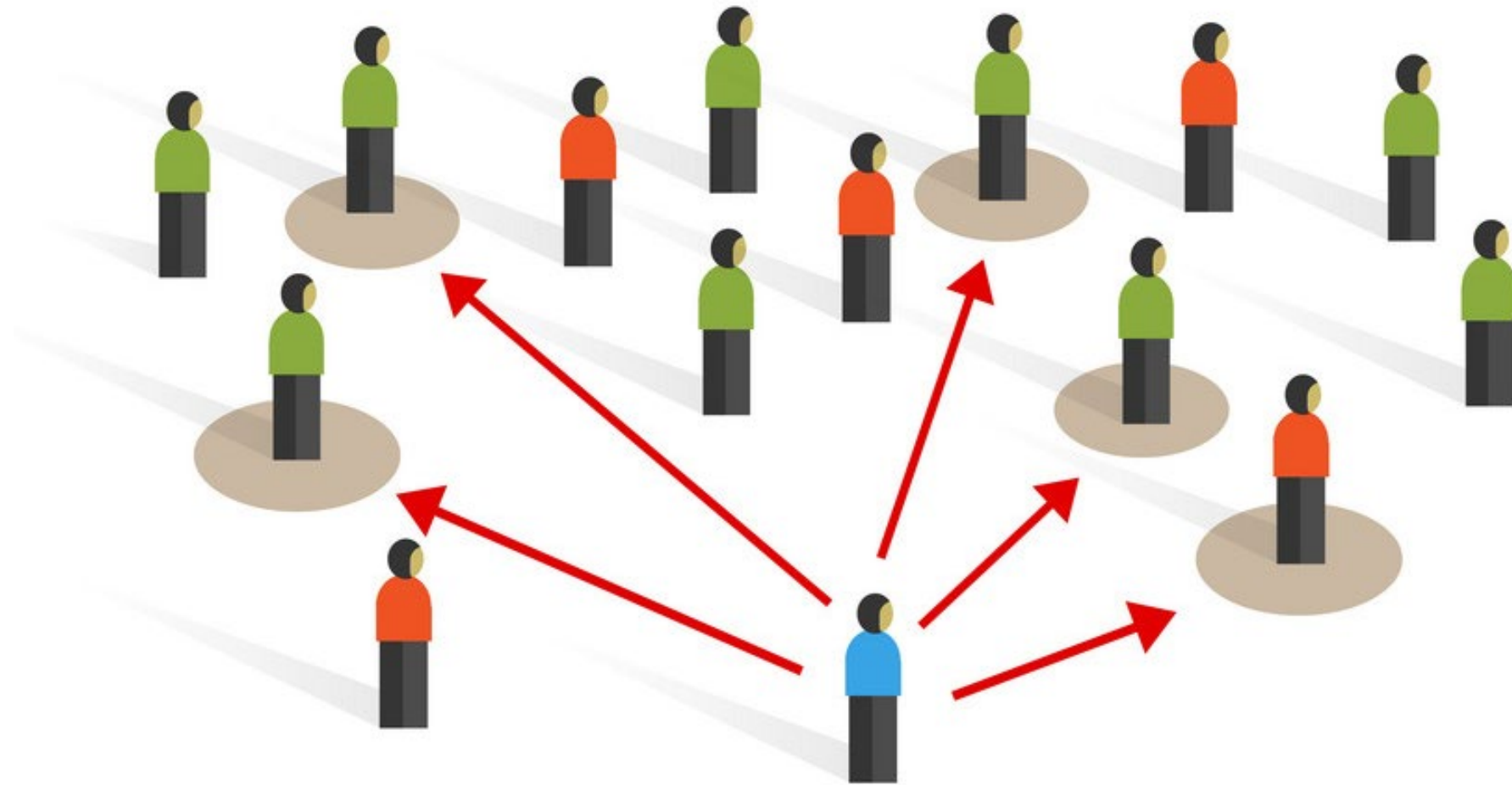
YEAH,
I LOST THEM OVER
THERE BUT THE
LIGHT'S BETTER HERE

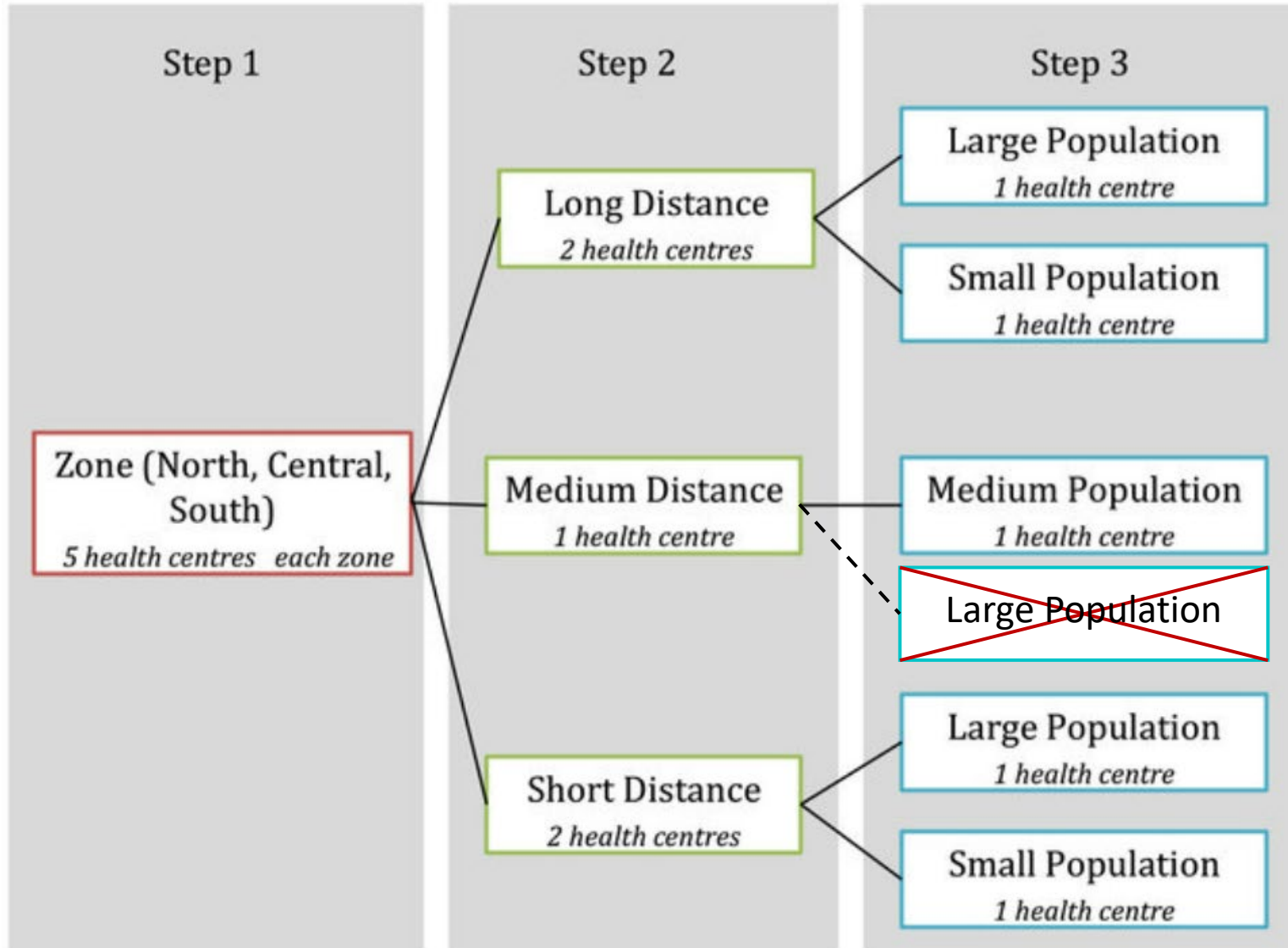
SAMPLING BIAS



" WE RECEIVED 500 RESPONSES AND
FOUND THAT PEOPLE LOVE RESPONDING
TO SURVEYS "

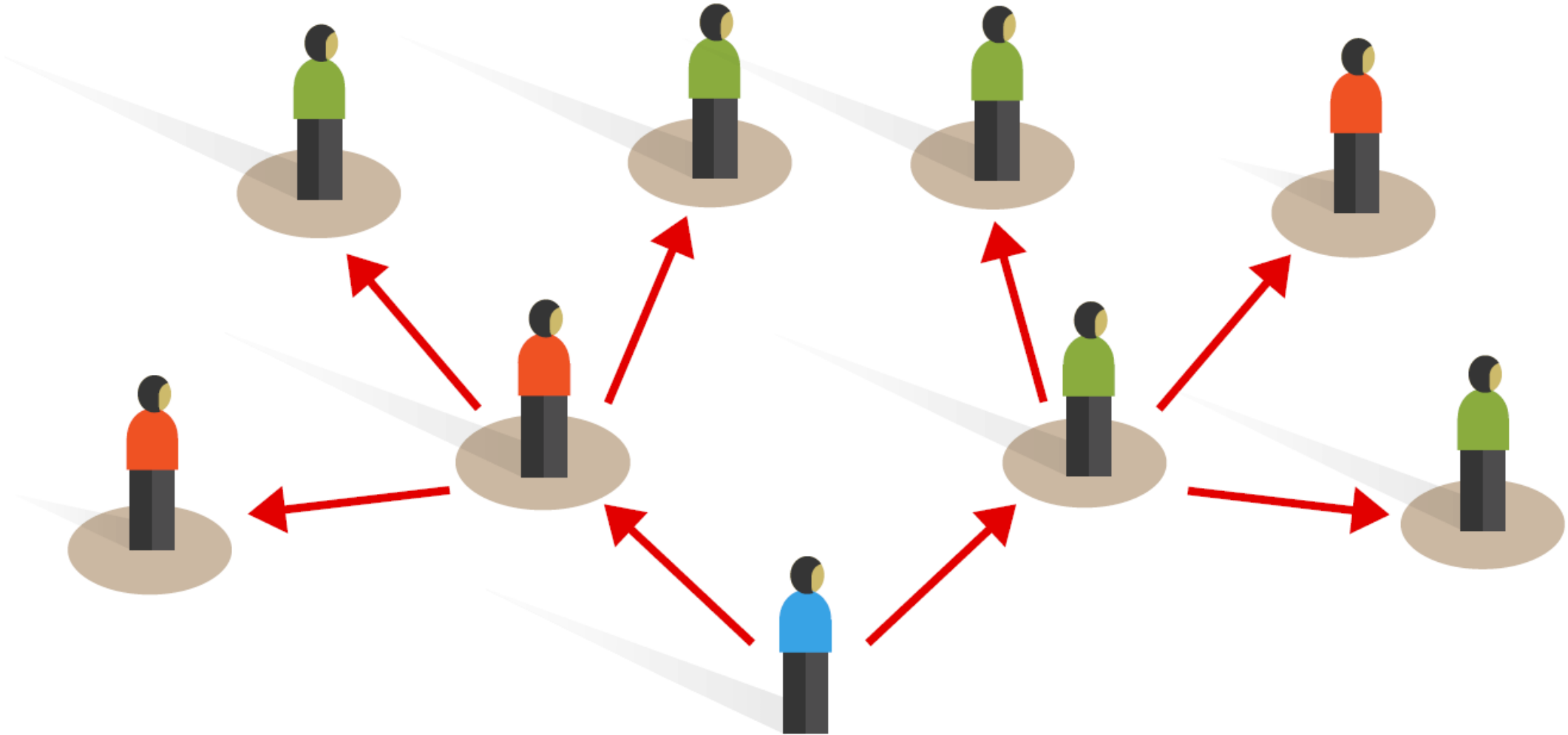
Purposive sampling





Some elements of the population have no chance of selection ($p = 0$)

Snowball sampling



The snowball effect

The snowball effect



What is this?

The Spread of Obesity in a Large Social Network over 32 Years

Nicholas A. Christakis, M.D., Ph.D., M.P.H., and James H. Fowler, Ph.D.

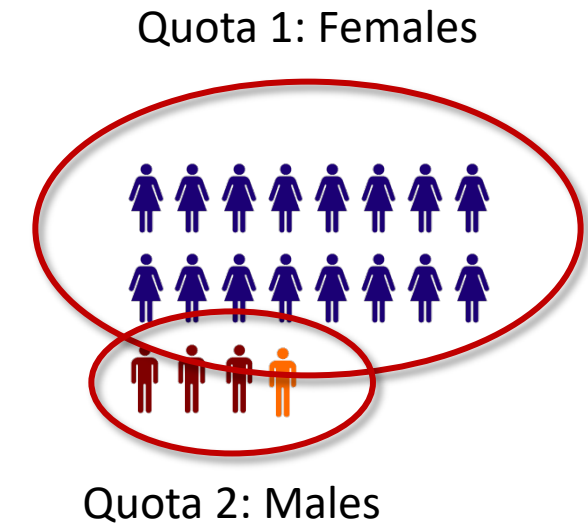
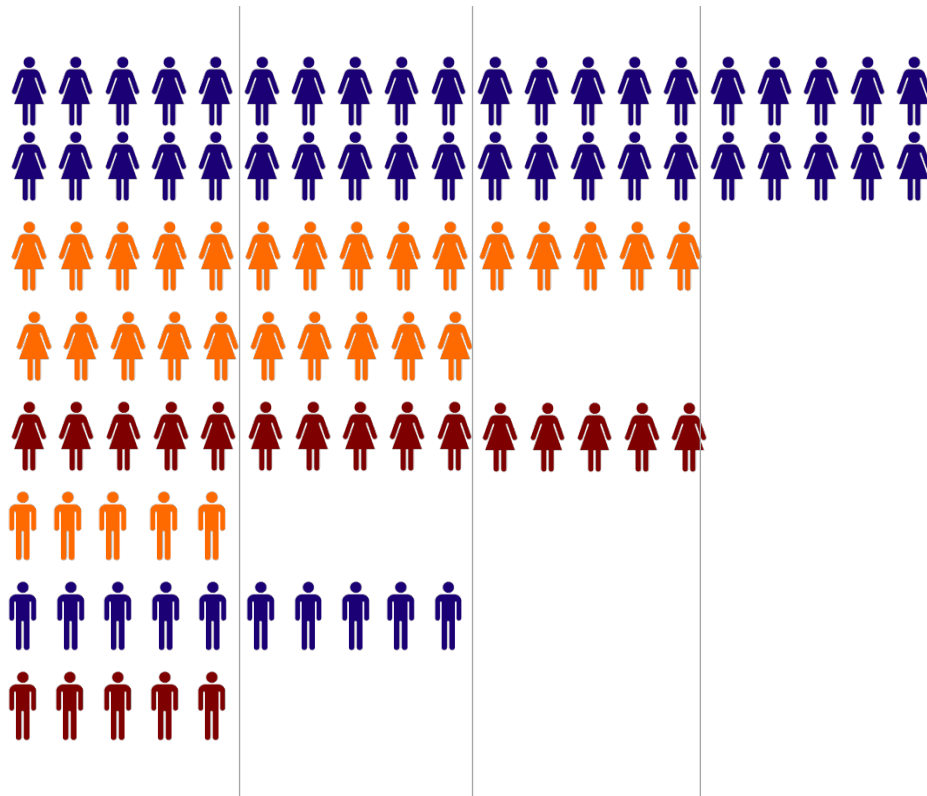
Video



1m48s

Quota sampling





1. Only the **selected** traits of the population are taken into account in forming the subgroups
2. Selection within subgroups is **not random**

Sacrifice is a step beyond convenience: A review of convenience sampling in psychological research in Africa

<https://sajip.co.za/index.php/sajip/article/view/1837/3207>

Aims and objectives

- What are the current use and reporting practices of convenience sampling in African psychology journals?
 - Categorise method-relevant sections of empirical research reports to establish the use of current convenience sampling method and reporting practices in African psychology journals.
 - Establish the methodological standards of convenience sampling in African psychology journals.

The study itself

- Study design
- Unit of analysis
- Target population
- Sampling strategy
- Sample size
- Generalisability of results

The results

- Use
- Reporting practices



05:00

mins: secs: type: ▼



Some results

TABLE 1: Frequency of convenience sampling in African psychology journals.

Journal	f	Reported
<i>Journal of Psychology in Africa</i> (JPA)	87	63
<i>South African Journal of Industrial Psychology</i> (SAJIP)	30	28
<i>South African Journal of Psychology</i> (SAJP)	22	12

f , frequency.

TABLE 2: Frequency of convenience sampling in method use.

Method	f	Reported
Quantitative	109	80
Qualitative	26	20
Combined	2	1
Mixed	2	2

f , frequency.

Some results

TABLE 3: Reported convenience sampling method characteristics.

Sampling method characteristics	<i>f</i>
Size reported	139
Sample justification	21
Sample strengths or weaknesses	14
Sample response rate	22
Generalisability	94
Representativeness	33
Information of influence of sample on results	51
Limitation statement (sample size)	50
Limitation statement (convenience sampling method)	43
Encourage future sampling from different contexts	78

f, frequency.

Some results

TABLE 4: Convenience sample participants' characteristics.

Sample characteristic	Total participants <i>f</i> = 44 678
Gender (<i>n</i> = 135)	
Male	19 307
Female	24 093
Age (<i>n</i> = 109)	
Age mean reported	59
Oldest participant	75
Youngest participant	8
Race (<i>n</i> = 80)	
Black (African) people	66
White (Caucasian) people	47
Mixed-race (coloured) people	33
Indians	23
Asians	17
Others	11
Missing	3
Language (<i>n</i> = 43)	
Data collected in own language	12
Data collected in English	32
Sample home language	
Occupation or employment	87
Education	87
Socio-economic status	14

TABLE 4 (Continues...): Convenience sample participants' characteristics.

Sample characteristic	Total participants <i>f</i> = 44 678
Philippines	1
Vietnam	1
United Kingdom	1
West Africa	1
Indonesia	1
Source detail (<i>n</i> = 135)	
University or college	43
Region or urban or rural areas	32
Organisation	32
Hospital or treatment centre	12
School	12
Festival or museum	3
Governmental institute	3
Jail	2
Online	2
Village	2
Venue	1
Attendance register	1
Orphanage	1

Some results

...more studies included statements on the generalisability of their results than their representativeness. These types of accounts were often under the limitation and recommendation sections of articles. Keyser and Duvenhage (2019) present an example of this statement: 'Limitations of this study include that the sample was drawn from among university students rather than the general population. This limits the generalisability of findings'

In addition to the generalisability of samples, some researchers also provided information on their sample's possible influence on their study results. Kagee, Bantjes, Saal and Sefatsa (2019), for example, state that because of the self-selection nature of their study, participants who experienced emotional distress might have been more prone to enrol for data collection than those who did not, in their study of posttraumatic stress disorder (PTSD) related to HIV testing.

Non-probability does not mean necessarily non-representativeness



Available online at www.sciencedirect.com

SCIENCE @ DIRECT®

Drug and Alcohol Dependence 73 (2004) 33–40



www.elsevier.com/locate/drugalcdep

The external validity of results derived from ecstasy users recruited using purposive sampling strategies

Libby Topp, Bridget Barker*, Louisa Degenhardt

National Drug and Alcohol Research Centre, University of New South Wales, Sydney, NSW 2052, Australia

Received 12 May 2003; accepted 2 September 2003

“than the sample of recent users. The similarities between the demographic and drug use characteristics of the samples are striking, and suggest that, at least in NSW, purposive sampling that seeks to draw from a wide cross-section of users and to sample a relatively large number of individuals, can give rise to samples of ecstasy users that may be considered sufficiently representative to reasonably warrant the drawing of inferences relating to the entire population. These findings may partially offset concerns that purposive samples of ecstasy users are likely to remain a primary source of ecstasy-related information.”

Sampling strategies for ‘special’ populations

‘Hidden’

‘Mobile’

‘Hard-to-reach’

no sampling frame

Active avoidance of
being found

**SPECIAL SAMPLING
METHODS**



**SPECIAL ANALYSIS
METHODS**

Sampling strategies for ‘special’ populations

Snowball sampling

Snowball sampling entails identifying an initial number of subgroup members from whom the desired data are gathered and who then serve as ‘seeds’, or study staff recruited respondents, to help identify other subgroup members (i.e. individuals who engage in the same types of behaviours) to be included in the sample.

Targeted sampling

Targeted sampling extends the ideas of snowball sampling to include an initial ethnographic assessment aimed at identifying the various networks or subgroups that might exist in a given setting. The subgroups so identified are then treated as sampling strata, and quota samples are chosen within each stratum using systematic sampling when feasible.

Time-location sampling

It takes advantage of the fact that some hidden populations tend to gather or congregate at certain types of locations. In time-location sampling (TLS), such sites are enumerated in a preliminary ethnographic mapping or presurveillance assessment exercise; the list of sites so developed is used as a sampling frame from which to choose a probability sample of sites, and data are gathered from either all or a sample of subgroup members found at the site during a pre-defined time interval (e.g. a randomly chosen 3-h time period on a randomly chosen day of the week).

Respondent-driven sampling

The method is similar to snowball sampling in that it involves chain referral sampling. However, the recruitment process is implemented in a manner that allows for the calculation of selection probabilities, thus it qualifies as a probability sampling method

(adapted from: Magnani et al. 2005)

Summary questions

- Define probability and non-probability samples
- Describe alternative sampling strategies and cite main advantages/disadvantages of each of them
- Sampling strategies and external validity
- Sampling strategies and data analysis
- Sampling strategies for special populations

References

- Aday, LA, and LJ Cornelius. 2006. *Designing and Conducting Health Surveys: A Comprehensive Guide. 3rd Ed*2006. San Francisco, CA: Jossey-Bass. Chapter 6.
- Fowler Jr, Floyd J. 2013. *Survey Research Methods*. Sage publications. Chapter 3.
- Scholtz, Salome E. 2021. 'Sacrifice Is a Step beyond Convenience: A Review of Convenience Sampling in Psychological Research in Africa'. *SA Journal of Industrial Psychology* 47 (0): 12. <https://sajip.co.za/index.php/sajip/article/view/1837>
- Magnani, Robert, Keith Sabin, Tobi Saidel, and Douglas Heckathorn. 2005. 'Review of Sampling Hard-to-Reach and Hidden Populations for HIV Surveillance'. *AIDS* 19 (May): S67. <https://doi.org/10.1097/01.aids.0000172879.20628.e1>.

Other resources

- Cornesse, Carina, and Michael Bosnjak. 2018. 'Is There an Association between Survey Characteristics and Representativeness? A Meta-Analysis'. *Survey Research Methods* 12 (1): 1–13. <https://doi.org/10.18148/srm/2018.v12i1.7205>.
- Topp, Libby, Bridget Barker, and Louisa Degenhardt. 2004. 'The External Validity of Results Derived from Ecstasy Users Recruited Using Purposive Sampling Strategies'. *Drug and Alcohol Dependence* 73 (1): 33–40. <https://doi.org/10.1016/j.drugalcdep.2003.09.001>.
- Heckathorn, Douglas D. 2011. 'Comment: Snowball versus Respondent-Driven Sampling'. *Sociological Methodology* 41 (1): 355–66.
- Johnson, Timothy P., Ananda Mitra, Ramona Newman, and John Horm. 1993. 'Problems of Definition in Sampling Special Populations: The Case of Homeless Persons'. *Evaluation Practice* 14 (2): 119–26. <https://doi.org/10.1177/109821409301400201>.
- Spreen, Marinus. 1992. 'Rare Populations, Hidden Populations, and Link-Tracing Designs: What and Why?' *Bulletin of Sociological Methodology/Bulletin de Méthodologie Sociologique* 36 (1): 34–58. <https://doi.org/10.1177/075910639203600103>.



Thank you

Annibale Cois

Education Building, 4th floor, Room 4060B

acois@sun.ac.za